

Keep the Angle

A Universal Geometry-Preserving Basis in Spectral Embeddings

the angular coordinate, commute-time degeneracy, and observer coarse-graining

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Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [8]. We show the geometry does not vanish—it migrates entirely into the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude $\times$ direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [5], previously justified heuristically; we give it a geometric reason. We **prove** (via von Luxburg–Radl–Hein) that the radial coordinate must degenerate to a local-degree quantity for intrinsic dimension $d \geq 2$, and **conjecture**, with strong empirical support, that the angular coordinate stays bi-Lipschitz to geodesic distance—flat in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$) fidelity obeys $\rho \approx 1.09 - 0.157d$ over that range (Spearman -0.881). The same scale-invariant-direction operation appears in KV-cache compression; a bounded-observer interpretation (Wolfram’s emergent geometry) is developed, and bounded, in the discussion.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [2]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [8]. Yet the most successful spectral-clustering algorithm [5] *row-normalizes* the embedding—projects each node to the unit sphere—and works well, with the normalization justified largely by heuristic. **This paper asks what row-normalization recovers, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s theorem kills; the *direction* carries the graph’s geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so it is not a heuristic but the geometrically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This “keep the scale-invariant direction” operation recurs beyond clustering—in KV-cache compression (PolarQuant transfers the scale-invariant direction) and in preference geometry (a scale-invariant “aversion angle” that transfers across decision domains). It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [11], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated.

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a proven Radial-Degeneracy *theorem* and an empirically-supported Angular-Preservation *conjecture*; (2) a measurement with controls (§5); (3) universality across five substrate types (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); and (5) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [2] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [8] proved the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/d_i + 1/d_j$; we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [5] row-normalize the eigen-embedding—the angular projection we study; we supply the geometric reason it recovers geodesics. **Emergent geometry and observer theory.** In the Wolfram model [9, 10, 3] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [11]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [4, 7, 6]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The commute-time embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$. Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$.

Intrinsic dimension. Throughout, d is the intrinsic dimension of the substrate: the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4.1 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime $d \geq 2$ refers to this measured intrinsic dimension; §5.4 confirms the $d = 1$ boundary behaves as predicted.

Theorem 1 (Radial Degeneracy). *For a graph of intrinsic dimension $d \geq 2$, as $n \rightarrow \infty$ the commute distance degenerates to an additively separable degree function [8], and the radial coordinate satisfies $r_i \rightarrow 1/\sqrt{d_i}$ (d_i the node degree). The radius is therefore geometry-free—a local-density coordinate carrying no geodesic information.*

Proof (sketch). A direct corollary of the von Luxburg–Radl–Hein degeneracy theorem: for $d \geq 2$ the resistance distance $R(i, j) \rightarrow 1/d_i + 1/d_j$, an additively separable form whose only per-node

content is the self-term, which is exactly r_i^2 . Verified numerically (§5.5): $\rho(r_i, 1/\sqrt{d_i}) = 0.92\text{--}0.99$ and radius-only geodesic preservation $\rho \leq 0.08$. $\square$

Conjecture 1 (Angular Preservation). *Row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Theorem 1, and the resulting direction of the low-mode eigenmap remains bi-Lipschitz to geodesic distance uniformly in the mode-count m and size n .*

Status. Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Theorem 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88\text{--}0.90$) even on quasi-1D graphs where Theorem 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i .

Dimension gating. Theorem 1 is a $d \geq 2$ phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Theorem 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d . **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
5	0.761	0.926	0.025
10	0.650	0.913	0.035
20	0.550	0.891	0.028
40	0.517	0.928	0.034

The angle carries the geometry (≈ 0.92 , flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2).

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only ≈ 1.5 D manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle- $\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

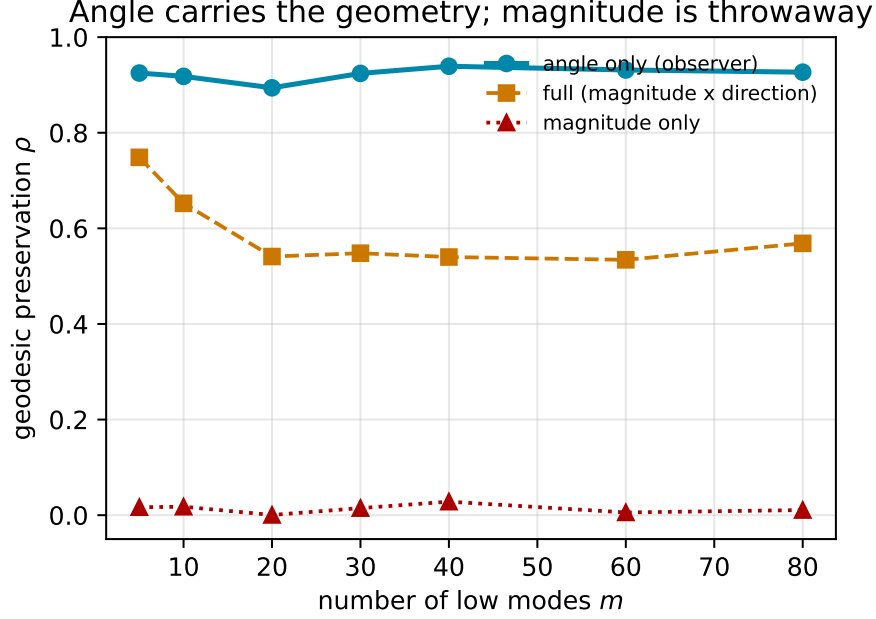


Figure 1: Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m .

5.3 Universality across substrates

substrate	meas. d	angle- ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0	0.45	0.00	—

Substrates with genuine low- d Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. The swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph rewriting or cultural sequential data.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$),

$$\text{angle-}\rho \approx 1.09 - 0.157d, \quad \text{Spearman} = -0.881, \quad d \in [1.0, 3.6].$$

The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \rightarrow 0$ fidelity, and the slope may vary with substrate roughness. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

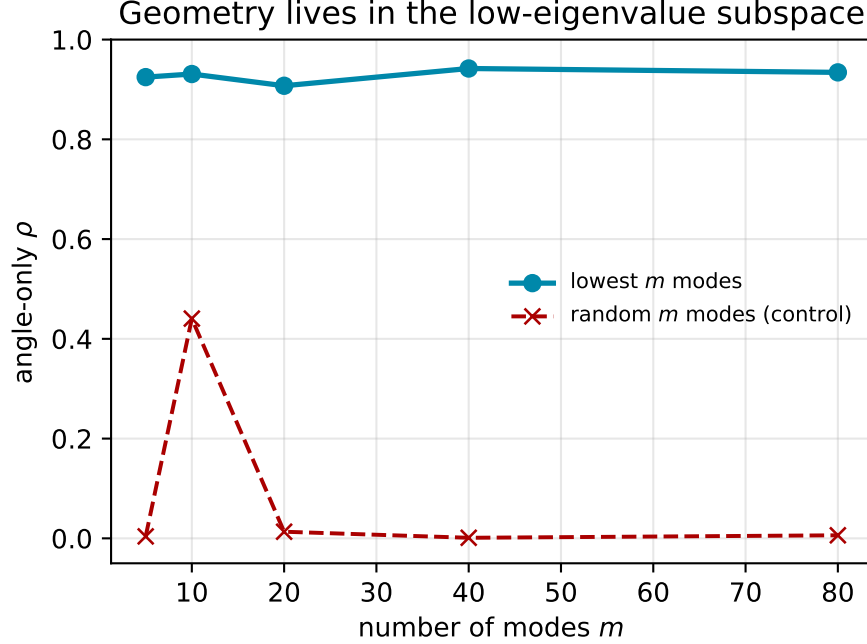


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not.

5.5 Verifying Theorem 1 and Conjecture 1

angle- ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{d_i}) = 0.92\text{--}0.99$; radius-only $\rho \leq 0.08$). Theorem 1 holds cleanly at $d = 3$ ($\rho(R, 1/d_i + 1/d_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an m -sweep at $n = 4000$, angle- ρ is flat ≈ 0.93 across $m = 5\text{--}80$ while the full embedding decays $0.773 \rightarrow 0.437$.

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geometric justification it previously lacked,

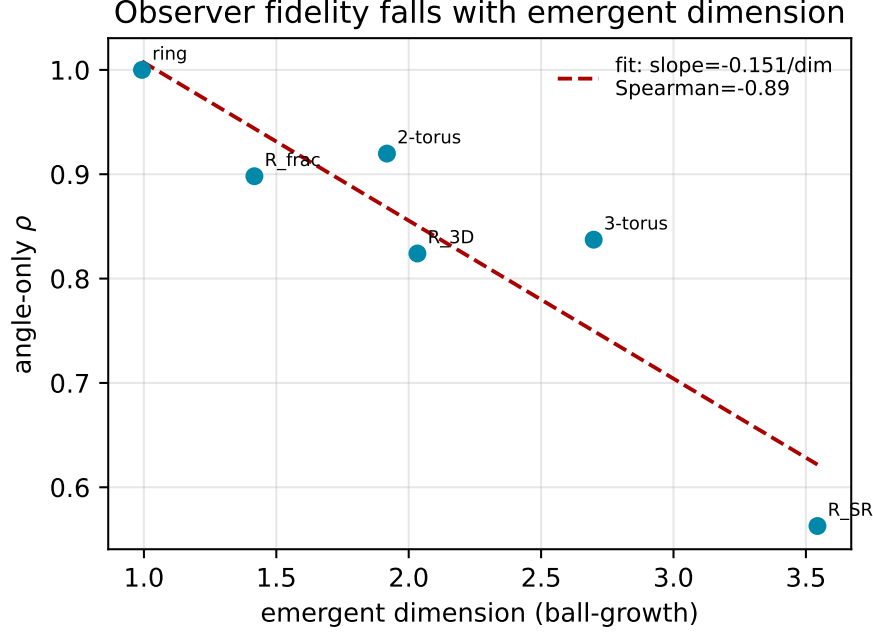


Figure 3: Angular fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules.

unifies it with the scale-invariant direction of KV-cache compression, and holds *universally* across substrates with genuine low- d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and cultural-sequential substrates is evidence that scale-invariant angular coarse-graining is a general law of how relational data carries geometric structure.

One interpretation is observer-theoretic. In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy-responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1; synthesizing clean manifolds above $d \approx 2$; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

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